RehabSense

A wearable sEMG + IMU sleeve for home-based stroke rehabilitation

CSE/EE 475 — Embedded Systems Capstone / EE 542 • Team 1 • Autumn 2026
Qian (Andy) Cai • Eyerusalem Earl • Rusha Maiti • Aidan Goodman
Instructor: Prof. Shwetak N. Patel • TAs: Alex Ching, Guilin Hu
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1 Summary

RehabSense is a low-cost wearable sleeve for home-based stroke rehabilitation that measures muscle activity and limb motion at the same time, grades each exercise repetition on the device, and tells the patient the result through a vibration on their own arm. A surface-EMG channel on the biceps and two inertial sensors on the upper arm and forearm let the system distinguish a repetition genuinely performed by the target muscle from one completed by shoulder elevation, trunk lean, or momentum — a distinction that motion-only rehabilitation devices cannot make and that no one is present to make at home. Each repetition receives a 0–100 score built from range of motion, tempo, and smoothness, together with a fatigue indicator derived from the EMG frequency spectrum. A second channel on the triceps arrives with the custom PCB and adds co-contraction as a fourth scored component. Results stream over Bluetooth Low Energy to a host application for the patient and, eventually, their therapist. The quarter delivers a working wearable prototype on a custom PCB, demonstrated live.

Document status. This revision incorporates measurements from a working prototype exercised on a volunteer on 21 October (ESP32 DevKit, 2 × BMI270, MyoWare 2.0, DRV2605L driving an ERM actuator). Findings from that session are tagged [21 Oct test] throughout, distinguishing measured values from assumed ones.

What changed in v1.5. Four decisions taken on 22 October, after the firmware had been brought into line with v1.4:

- **One EMG channel for the prototype.** A single MyoWare 2.0 on the biceps — two electrodes on the muscle belly plus one reference at the elbow, three in all. The triceps channel, and with it the co-contraction score, arrives with the custom PCB.
- **The ERM is the production actuator.** The move to an LRA, proposed in v1.4, is reversed; every LRA reference is removed.
- **A soft click at 80% of the target angle** is added — an open question in v1.4, now settled yes.
- **A fatigue warning advises and does not end the set** — the other open question, settled the same way.

Implementation status. Firmware implementing this document is in the repository at `cse475/firmware`, with the design recorded in *Firmware Design v2.0*. Everything specified here is built: the activation-gated repetition cycle, grading, the seven-pattern haptic vocabulary with priority and preemption, the five-step calibration protocol, the full BLE command

set including runtime configuration, and backfill after a dropout.

Five test suites run on a laptop with no hardware, no radio and no ESP-IDF, and all five pass. The repetition suite compiles the real `rep_cycle.c`, `grading.c` and `activation.c` and drives sixteen scenarios through them, including a passively swung arm that counts nothing. What remains outstanding needs a bench, volunteer data, or a team decision; none of it is firmware work. The custom PCB is in design.

2 Problem statement

Stroke is among the most common causes of long-lasting upper-limb impairment. The CDC reports that more than 795,000 people in the United States have a stroke each year, that stroke is a leading cause of serious long-term disability, and that it reduces mobility in more than half of survivors aged 65 and above [1]. Recovery of arm function depends on sustained, high-repetition practice, the majority of which is prescribed as a home exercise programme and performed without supervision between clinical visits.

Unsupervised practice introduces two failure modes that neither the patient nor the therapist can currently observe.

- **Compensatory movement.** When the target muscle cannot generate sufficient force, the movement is completed by recruiting other structures — elevating the shoulder, leaning the trunk, or using momentum. The joint trajectory still appears complete, but the intended muscle performs little of the work. Because the substitution is repeated across many sessions, it is reinforced as a motor pattern and becomes progressively harder to correct.
- **Unmonitored fatigue.** Repetition quality degrades as the muscle fatigues, and fatigued practice contributes disproportionately to compensatory patterns. No objective indicator currently tells a patient when to rest or a therapist how much of a prescribed session was performed under fatigue.

Both failure modes are invisible in the dominant documentation method — a self-reported repetition count on paper — so the clinician returns to a session with no information about how the preceding weeks of practice were actually performed.

Existing approach	Limitation for unsupervised home practice
Paper home-exercise programmes	No feedback mechanism; adherence and movement quality are both unrecorded.
Camera- and IMU-based rehabilitation games	Measure kinematics only. Cannot determine whether the target muscle was recruited, cannot observe fatigue, and score compensated movements as successful.
Clinic-based rehabilitation robots and exoskeletons	Accurate and well validated, but expensive, non-wearable, and dependent on travel to a facility.
Therapist-supervised EMG biofeedback units	Observe muscle activity accurately, but are stationary, typically single-channel, and are not designed to score exercise repetitions.

Gap addressed by this project. Kinematic measurement alone can be satisfied by a compensated movement; electromyography alone cannot localise where a movement originated. No low-cost wearable system currently measures both simultaneously and returns feedback to

the patient within the session in which the error occurs.

Target applications. Home rehabilitation with real-time movement-quality feedback; tele-rehabilitation, in which objective session data is reviewed remotely; a clinic-to-home bridge that records progress between appointments; and a low-cost research platform for rehabilitation studies.

3 Deliverables and critical features

3.1 What will be demonstrated on 10 December

1. A wearable sleeve carrying one EMG channel, two IMUs, and a haptic actuator, driven by a custom PCB in a 3D-printed pod with its own battery, power switch, user button, and status LED.
2. A live session: don the sleeve, pass the contact check, run calibration, and perform a set of repetitions with the counter advancing and a haptic response on each one.
3. Deliberate compensation — shoulder elevation — detected live, flagged with a distinct haptic pattern, reflected in a lowered score, and named on the host display.
4. A passively swung arm rejected: no repetition starts, because the biceps never activates.
5. A fatiguing set, with the median-frequency trend falling and the rest cue firing.
6. Exercise parameters — hold duration, repetition target, target angle — changed live from the host application.
7. The host application showing live traces, per-repetition scores, and a session summary.

3.2 Critical features

Feature	Priority	Notes
Single-channel sEMG acquisition	Must	Biceps; raw, band-limited. The triceps channel arrives with the custom PCB
Dual-IMU elbow angle	Must	Relative orientation, 5–10° accuracy
On-device repetition detection	Must	Deterministic; gated on biceps activation
On-device grading, 0–100	Must	Three equal-weight components; a fourth with the second channel
Compensation detection	Must	Shoulder elevation, trunk lean, momentum
Haptic feedback, three outcomes	Must	Clean, flagged, set complete; distinct by rhythm
Movement waypoint cue	Should	Soft click at 80% of the target angle
Fault and rest cues	Must / Should	Fault is a must (contact loss, sensor error, low battery); the fatigue rest cue is a should
Non-overlapping cue playback	Must	Single-slot priority queue; a fault preempts a playing cue
Driver configuration self-test	Must	Register read-back before every session
Host pattern test	Should	Named pattern fired on demand for bench and volunteer testing
BLE streaming and control	Must	Live features, events, switchable raw stream
Host application	Must	Python for the prototype; Flutter thereafter
Calibration under 30 s	Must	Six prompted steps, individually repeatable
Fatigue detection	Should	Median-frequency regression across the set; advises, does not end the set
Custom PCB	Should	Development board is the fallback
Battery operation, 2 h	Should	LiPo with fuel gauge
Enclosure and sleeve integration	Should	3D-printed pod
Therapist dashboard	Could	Backend topics exercised; UI is later work

4 Scope and phasing

Phase 1 — rapid prototype, due on the project website 29 October, noon

A functional benchtop prototype built on the ESP32-S3 development board with a MyoWare 2.0 sensor, demonstrating the complete signal path: one EMG channel and two IMUs acquired on a shared timebase, elbow angle derived to within 5–10°, repetitions detected and graded on-device, the correct haptic pattern issued, and the full stream logged by the Python host over BLE.

Excluded from this phase: the custom PCB, the sleeve assembly, the enclosure, the patient-facing game, and electrode-site optimisation.

Phase 2 — wearable prototype, 29 October to 10 December

Custom PCB designed, fabricated, and brought up; sensors integrated into a compression sleeve; 3D-printed pod with power switch, button, and RGB status LED; grading parameters tuned against recorded volunteer data; two hours of battery operation; and a demonstration script that runs reliably twice in succession.

The MyoWare 2.0 module is the sensing front end for Phase 1 only. The custom PCB replaces it with an integrated two-channel analogue front end, taking the electrode count from three to five: the triceps pair is added and the single driven reference is shared. Firmware is written against an abstract acquisition interface with both backends in place, so the change does not propagate above the driver layer, and everything that needs a second channel is already written and guarded on one constant rather than waiting to be added.

5 System design

Design summary. A wearable sleeve measures muscle activity and limb motion concurrently, determines on-device whether each repetition was performed correctly, and communicates the result to the patient through a vibration pattern delivered at a predictable point in the movement.

5.1 Sensor and electrode placement

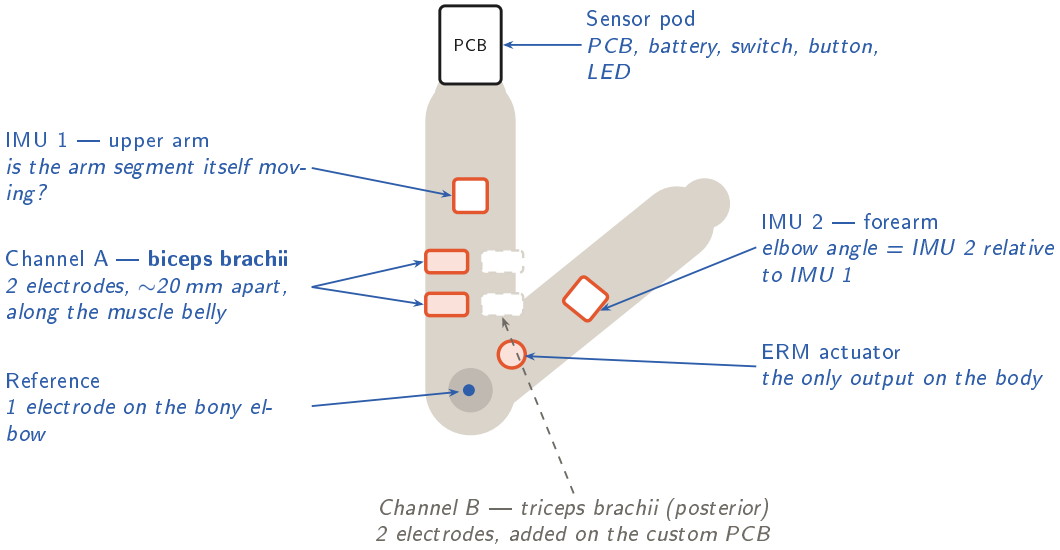


Figure 1: *Sensor and electrode placement. The Phase 1 prototype uses three electrodes: two on the biceps belly plus one reference at the elbow. The custom PCB adds the triceps pair shown dashed, sharing the same driven reference, for five in all.*

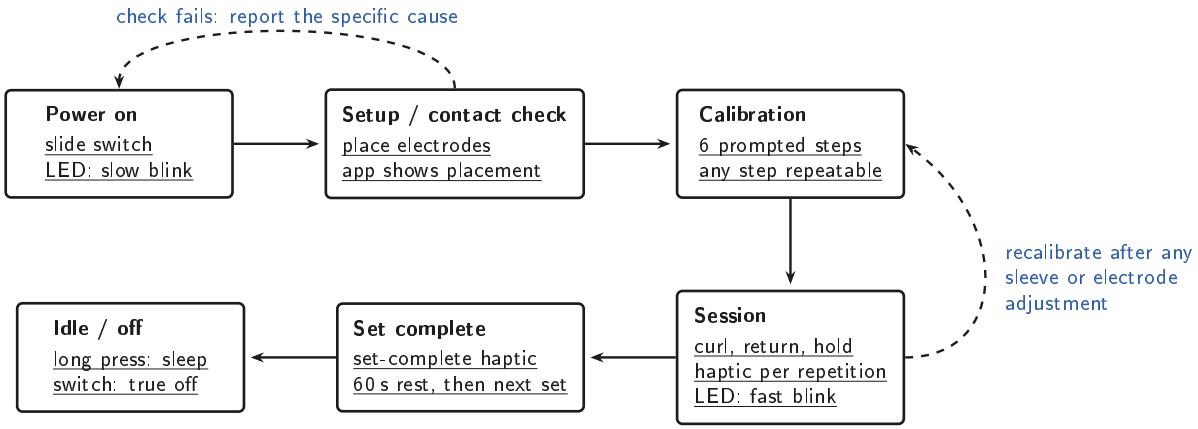
Site	Role	Electrodes
Biceps brachii	Agonist. Gates repetition start, and supplies effort and fatigue measures.	2 on the muscle belly, ~20 mm apart, aligned with the muscle fibres, per SENIAM guidance [4]
Elbow (bony landmark)	Reference	1
Triceps brachii	*Antagonist. Supplies the co-contraction score. Phase 2, with the custom PCB.*	*2 on the posterior upper arm, ~20 mm apart*

Rationale for measuring the biceps first. The biceps brachii is the agonist during elbow flexion, and its activation gates the start of a repetition — which is what allows a passively swung arm to be rejected before it is ever counted. That single gate is the whole argument of §2, so if only one channel can be fitted, this is the one.

The triceps is the antagonist, and simultaneous activation indicates co-contraction: a distinct impairment that kinematic sensing cannot detect. It is worth measuring and it is not worth delaying the prototype for, because co-contraction grades a repetition that the biceps channel has already decided to count. It is therefore deferred to the custom PCB, which can carry it without a second MyoWare module, a second reference electrode, or two more leads on the sleeve.

5.2 Device states and session flow

State	Entered when	Behaviour
Idle	Boot, or session ended	BLE advertising; LED slow blink
Calibrating	Calibrate command, or double button press	Step prompts over BLE and by haptic
Session	Start command, or short button press	Acquisition, grading, haptics, streaming; LED fast blink
Charging	USB connected	Measurement disabled; the analogue rail is cut in hardware
Fault	Contact loss, sensor error, or low battery	Session paused and saved. The fault cue plays first; haptics are muted through the driver enable pin only after it finishes, about 900 ms later, so the fault state cannot silence its own warning



Boxes are states the patient can be in; underlined items are the actions and indicators available in that state. The patient's on-body interface is the switch, the button, the LED, and the actuator; the host application provides the visual view.

Figure 2: *End-to-end session. A default prescription is three sets of twenty repetitions with sixty seconds of rest between sets.*

5.3 Calibration

Calibration runs as a sequence of short, separately commanded steps, so a failed step can be repeated on its own without restarting the protocol. Total capture time is approximately sixteen seconds on the single-channel prototype, or about twenty-five including prompts — against a Must of under

thirty in §3.2. The triceps step returns with the second channel and adds nine.

Step	Time	Patient action	Captured, and pass condition
0. Setup	—	Sits; sleeve on; electrodes placed using the app's placement images	Contact check: no channel saturated, mains pickup within limits, both IMUs responding. Failure reports the specific cause ("press the electrode firmly") with the fault cue.
1. Rest	5 s	Arm relaxed at the side, completely still	EMG resting mean and standard deviation, which set the muscle-on threshold, and gyroscope bias for both IMUs. Rejected and repeated if movement is detected — movement here corrupts the bias, and therefore the joint angle, for the whole session with no obvious symptom.
2. Straight arm	2 s	Straightens the arm *as far as is comfortable* and holds still	The 0° reference pose, and the neutral-pose quaternions for both IMUs. Rejected if not still during capture.
3. Biceps maximum	3×3 s	Elbow at ∼90°, pushes up against a table and contracts the biceps	Biceps 100% reference, best of three attempts, each taken as the strongest 500 ms window. Must be clearly above rest; if not, front-end gain is raised and the step repeated.
4. Triceps maximum	3×3 *s*	*Same posture, pushes down*	*Triceps 100% reference, by the same method. Required by the co-contraction score; Phase 2, with the second channel.*
5. Range sweep	∼6 s	Two slow, full-range curls	Flexion axis, taken as the principal axis of the relative rotation over the sweep, so sleeve alignment is not assumed; achievable $\theta_{\min}$ and $\theta_{\max}$; baseline median frequency.

Findings incorporated from the 21 October session.

- **[21 Oct test]** Step 2 must instruct "comfortable straight," not "locked straight." Calibrating against a hyperextended pose caused returns to finish 10–12° short of the finish threshold, so repetitions did not complete.
- **[21 Oct test]** Step 4 was added because the co-contraction score normalises the triceps channel as well and therefore requires its own reference. With one channel there is no triceps to normalise, so the step is deferred rather than dropped — the reasoning survives, and returns with the hardware.
- **[21 Oct test]** A stillness check gates acceptance of steps 1 and 2. The rest step also establishes gyroscope bias, and movement during it corrupts the joint angle for the remainder of the session.
- **[21 Oct test]** Recalibration is required after any electrode or sleeve adjustment. In the session log, resting EMG read approximately 40% of the calibrated maximum after the hardware shifted, which invalidates both the activation scale and the "relaxed between repetitions" check.

Maximum contraction on an impaired limb. Maximal effort is inconsistent on a paretic limb, and a single poor attempt mis-scales every subsequent measurement. Three mitigations apply: the measurement is the best of up to three attempts rather than a single trial; if the best attempt fails to exceed the rest baseline by a defined margin the device flags the reference as unreliable and scales against the patient's own stored history instead; and a clinician-supplied normalisation value, pushed from the host, supersedes the measured one.

Persistence. Results are written to NVS under a patient profile key and survive power cycles. Steps 1, 2, and 5 are re-run every session, since they reset gyroscope bias and IMU drift and absorb day-to-day variation in sleeve placement. Steps 3 and 4 may be reused from the stored profile until it expires or the patient requests recalibration.

5.4 Repetition detection

A repetition begins when the biceps activates and the arm rises past the start threshold, and is counted when the arm has returned below the finish threshold and been held still there for a configurable interval. Counting on the return means the score covers the whole movement, including the eccentric phase, and the confirming hold removes the ambiguity of counting on a threshold crossing alone.

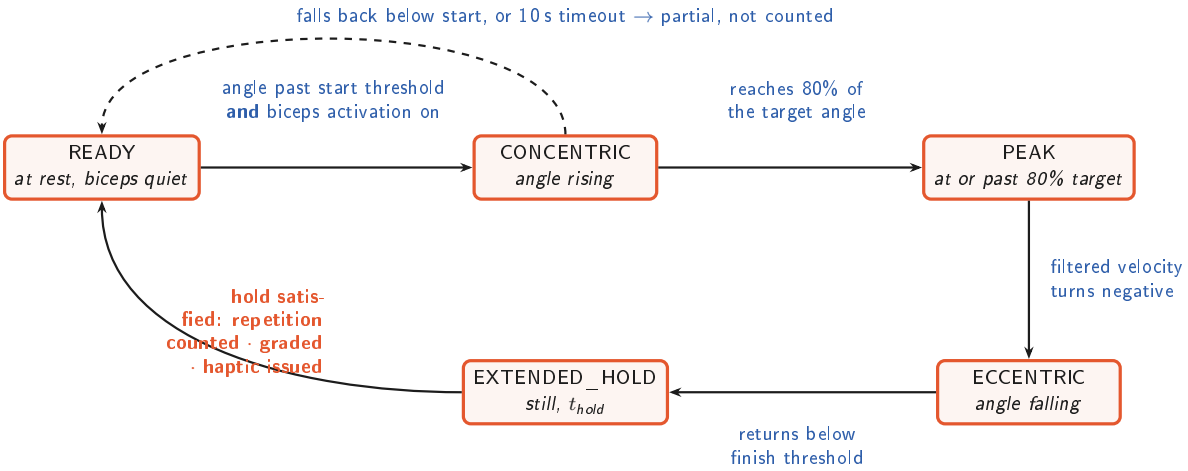


Figure 3: *Repetition cycle. Start is gated on biceps activation; the repetition is counted after the arm returns to the extended position and holds still there.*

Thresholds are derived from the calibrated range of motion and the therapist's target angle rather than from fixed anatomical angles. Counting remains traceable: every count corresponds to a full excursion with the agonist active, confirmed by a hold of defined duration. Sixteen scenarios pass in `test/test_rep.c` without hardware attached, among them a passively swung arm that counts nothing, a long hold at the top that counts nothing, set completion at the target, tempo scoring, and a shoulder hike that does not also register as trunk lean.

5.5 Grading and start-position checks

Each repetition receives a score from 0 to 100, reported to the host after every repetition. The patient receives only a pass/fail distinction, through the choice of haptic pattern; the numerical score is shown in the application.

Parameter	Default	Basis
Target angle	Therapist-set, 110°	[21 Oct test] Volunteer repetitions peaked at 89–98°. Adequate for testing; the therapist starts patients lower.
"Target reached"	80% of target angle	Defines the peak criterion.
Start threshold	15°	Configurable; 20° for patients who cannot fully extend.
Finish threshold	10° (5° hysteresis)	Configurable; 15° alongside a raised start threshold.
`hold_duration_ms`	1000 ms, range 300–3000	The confirming hold at the extended position. Host-tunable, because an impaired limb may not hold steadily for a full second.
`hold_velocity_thresh`	15°/s	A hold requires the limb to be stationary, not merely below the threshold.
Biceps activation on	Envelope above rest mean $+3\sigma$ for 50 ms; off below rest mean $+2\sigma$ for 100 ms	[21 Oct test] Replaces a fixed 10% threshold, and adapts to each session's own noise floor.
Direction reversal	Filtered velocity < 0, 5 Hz low-pass	[21 Oct test] A fixed -10°/s was used to avoid jitter.
Tempo window	Therapist-set, 1.5–3.5 s, start to finish	[21 Oct test] Volunteer repetitions ran 3.5–9 s. The confirming hold is excluded from the tempo measurement.
Repetition timeout	10 s without completion → partial	[21 Oct test] The session contained 16.6 s and 24.7 s "repetitions."
Session structure	3 sets × 20 repetitions, 60 s rest	Repetition goal is therapist-set and host-configurable.

Component	Weight	Measure, and status
Range of motion	1/3	Peak angle divided by the therapist's target angle; also reported against the healthy reference corridor. [21 Oct test] Working.
Tempo	1/3	Repetition duration, start threshold to finish threshold, against the prescribed window. [21 Oct test] Working.
Smoothness	1/3	Count of velocity peaks exceeding 10% of the repetition's peak velocity; a single peak scores 100. [21 Oct test] Working.
Co-contraction	*0*	*Overlap of biceps and triceps activation, evaluated only while the biceps is active. Needs the second channel and the triceps reference from calibration step 4; weighted zero until both exist.*
Deductions, applied after the mean; result clamped to 0–100		
Shoulder compensation	−20	[21 Oct test] Upper-arm IMU rotates more than ∼20° from its calibration pose during the repetition.
Trunk lean	−15	Common-mode rotation of both IMUs greater than 10°.
Momentum / arm swing	−15	Upper-arm angular velocity above 30°/s sustained for 200 ms during the concentric phase.

Components carry equal weight by default, and the score normalises by the weights actually in force — so a component weighted zero redistributes its share across the rest rather than capping the best achievable score below 100. Without that, a perfect repetition on the single-channel prototype would score 75, and the number on the therapist's screen would mean something different from the number in this document.

Weights and deductions are host-configurable (§5.9.1) so they can be tuned against volunteer data

without rebuilding firmware. Steadiness during the confirming hold is computed and reported but does not contribute to the default score; it is available as an optional fifth component.

Start-position and movement-quality checks

Condition	Detection	Response
Arm not returned near straight	Angle remains above the finish threshold	The hold never starts, so no repetition completes
Not relaxed between repetitions	Biceps envelope still above the activation threshold while at rest	Form cue; "relax your arm" shown in the app. **[21 Oct test]** The 40% resting EMG observed after a sleeve shift would trigger this.
Shoulder compensation	Upper-arm IMU rotated more than ∼20° from the calibration pose during the repetition	Repetition flagged; form cue
Passive swing	Angle rises with the biceps inactive	Repetition never starts
Sudden jolt	Acceleration spike on either IMU	That segment excluded from scoring

Three behavioural rules apply regardless of the weights. Every completed repetition is counted and then graded — no repetition is discarded silently, because a counter that fails to advance gives the patient no interpretable information. A repetition flagged for compensation still counts toward the repetition target: movement quality is expressed in the score, not in the counter. And compensation detected at any point during the movement is latched for that repetition and reported at the confirming hold, so feedback arrives at a single predictable moment rather than interrupting the movement.

Two defects identified by the test suite are recorded here because both would have manifested as implausible scores rather than as obvious failures. The smoothness term initially rewarded slower movement, since mean change in velocity scales with speed; it is now normalised by the movement's own mean speed. The steadiness term initially penalised a deeper curl, because the measurement window included the end of the ascent; the window now opens only once the limb has settled.

5.6 Haptic feedback

Feedback is delivered by a single eccentric rotating mass (ERM) actuator — the Adafruit 1201 — at the end of the confirming hold, when the patient is stationary. An ERM's vibration frequency is coupled to its amplitude, so "stronger" also means "faster" and cannot carry independent meaning. Patterns are therefore distinguished by *rhythm* — pulse count, width, and gap — and the six result cues are played at a single consistent amplitude, which is also easier for a patient to learn than a set of intensity gradations.

The DRV2605L's overdrive and active braking, applied from the stored calibration, sharpen the onset and shorten the decay enough to make the pulses and 70–80 ms gaps below perceptible as distinct events rather than one smeared buzz. Without a driver IC the gap floor would be roughly 120 ms and the vocabulary would have to be coarser.

Event	Pattern	Interpretation
Repetition counted, no fault	One sharp click, 100 ms	The repetition counted and met the form, range, and tempo criteria.
Repetition counted, form flagged	Three pulses, 80 ms on, 80 ms off	Counted, but incomplete range, outside the tempo window, or compensation detected.
Final repetition of the set	Two 70 ms pulses with 70 ms gaps, then one 500 ms pulse	The repetition target has been reached.
Rest now	One 1000 ms ramp up and down	Fatigue flag raised.
Fault	Two 300 ms pulses, 300 ms apart	Contact lost, sensor fault, or low battery.
Button press acknowledged	One 40 ms tick	Confirms every button press.
80% of the target angle	One 30 ms click at reduced level	You may begin the return. The only cue delivered mid-movement.

[21 Oct test] The 21 October notes identified a collision between a double-pulse "incomplete" cue and a double-pulse fault cue. Using three pulses for a flagged repetition resolves it and leaves the double pattern available for faults.

Each repetition receives exactly one of the first two patterns. On the repetition that completes the set, the set-complete pattern follows the per-repetition pattern after a 300 ms gap, so the patient receives both the quality result and confirmation that the set is finished.

Scheduling rules. One cue plays at a time. A fault preempts a playing cue and clears anything pending. A button-press tick is dropped if any cue is playing, as is the soft click — a waypoint the patient misses costs nothing, whereas a result they miss costs the repetition. Otherwise a new cue takes the single pending slot only if its priority exceeds that of the cue already waiting. Priorities, lowest to highest: tick and soft click, repetition result, set complete and rest, fault.

Timing. The haptic task runs at the highest application priority on a 1 ms scheduler tick, and each pattern segment is timed from a hardware timer so that pulse widths and gaps do not drift. The scheduler begins output on the same call that receives a request, so it contributes no latency of its own. Worst-case latency from the end of the confirming hold to perceptible vibration is approximately 43 ms, within the 50 ms budget.

Cue blanking. A cue's vibration couples into both the IMUs and the electrodes. The proposal is for firmware to suspend IMU and EMG analysis for the duration of each cue, so that a cue cannot be scored as a jolt or disturb the following repetition. *This needs agreement between the firmware and algorithms owners before implementation (§11.2).*

Host pattern test. The host can fire any named pattern — tick, clean, flagged, set complete, rest, fault — on demand, for bench verification and for the through-fabric discrimination test with volunteers.

The soft click, and why it is shaped differently. Settled on 22 October: a soft click fires at 80% of the target angle to indicate that the patient may begin the return. It is the only cue delivered while the patient is moving, and everything about it is chosen so that it cannot be mistaken for a

result — it is the shortest in the set, it plays at a reduced level, and it skips the overdrive kick that gives the other cues their snap. The six result cues keep their single consistent amplitude; the soft click sits deliberately outside that class rather than breaking the rule within it.

5.7 Hardware

Function	Phase 1 (29 Oct)	Phase 2 (custom PCB)
MCU	ESP32-S3-DevKitC-1 v1.1 (N8R8)	ESP32-S3 module on the PCB
sEMG	MyoWare 2.0 × 1, RAW output, 3 electrodes	Integrated two-channel analogue front end with driven reference, 5 electrodes
ADC	ESP32-S3 internal ADC1, continuous DMA, ∼9.5 effective bits	TI ADS131M02 over SPI, 24-bit, simultaneous sampling
Motion	2 × SparkFun BMI270 breakouts, SPI	2 × BMI270 on board, SPI
Haptics	Adafruit DRV2605L breakout + Adafruit 1201 ERM	DRV2605L + ERM on board
Power	USB from the development board	LiPo cell, charge circuit, fuel gauge, slide switch on the battery line
Indication	On-board WS2812	RGB status LED
User input	On-board BOOT button (GPIO0)	Dedicated button in the pod

Firmware is written against an abstract acquisition interface with both the internal-ADC and ADS131M02 backends in place, so moving from Phase 1 to Phase 2 is a single assignment at initialisation and touches nothing above the driver.

User button. Short press: start or pause the session. Long press (3 s): sleep. Double press: recalibrate. Every press is confirmed by a haptic tick.

Status indication. The PCB carries an RGB LED, so device states are colour-coded as well as blink-coded. A separate charge indicator is driven directly from the charger's status pin and requires no firmware.

State	Colour	Pattern
Idle, advertising	Blue	Slow blink
Connected	Blue	Solid
Calibrating	Amber	Pulsing
Session running	Green	Fast blink
Battery below 15%	Amber	Double blink every 2 s
Fault	Red	Solid

Below 15% charge the device completes the current set, then stops and saves rather than terminating mid-set.

Development pin assignment. The map below applies to the Phase 1 build. On the custom PCB, EMG acquisition moves to the ADS131M02 on SPI2 with its own chip select, and GPIO4, GPIO5 become available.

Signal	Pin	Peripheral	Note
EMG (biceps) RAW	GPIO4	ADC1_CH3	12 dB attenuation, ∼0–3.1 V window
EMG B (triceps) RAW	*GPIO5*	*ADC1_CH4*	*Reserved; unused on the single-channel prototype*
Battery sense	GPIO7	ADC1_CH6	2:1 divider, read while continuous mode is paused
SPI SCK / MOSI / MISO	GPIO12 / 11 / 13	SPI2 (FSPI)	10 MHz, shared by both IMUs
CS, upper-arm / forearm IMU	GPIO10 / 14	GPIO out	
I^2C SDA / SCL	GPIO8 / 9	I2C0	DRV2605L only, 400 kHz
Haptic PWM	GPIO6	LEDC	DRV2605L IN/TRIG
Haptic enable	GPIO15	GPIO out	Hard mute, bus-independent
Status LED	GPIO38	RMT	On-board WS2812
User button	GPIO0	GPIO in	On-board BOOT button

Pins avoided: GPIO19–20 (USB D−/D+), GPIO26–32 (SPI flash), and GPIO33–37 (octal PSRAM on the N8R8 module).

5.7.1 Haptic driver: I^2C configuration, PWM drive.

Power-up sequence. Reset the DRV2605L; leave standby; select ERM drive mode (`N_ERM_LRA` = 0 in Feedback Control, `0x1A`); set the rated voltage and overdrive clamp from the actuator datasheet; run auto-calibration; store the results in NVS; switch the mode register to PWM input; then read back every configured register. A mismatch fails the self-test before the session begins. If auto-calibration fails, the driver falls back to the stored calibration; if none exists, initialisation fails rather than running with unknown drive parameters. PWM cannot replace this sequence — it follows it.

PWM drive. LEDC at 20 kHz, 10-bit, with 50% duty as zero drive and deviation from 50% setting amplitude. The output is held at zero drive before the chip is switched into PWM mode, so the actuator cannot twitch during configuration. The driver's overdrive and braking continue to shape each pulse from the stored calibration while the PWM duty sets amplitude; this configuration is to be confirmed against the datasheet during bring-up.

Enable pin. Disabling holds EN low after first commanding zero drive. Re-enabling re-sends the full configuration unconditionally, because the chip loses its registers while EN is low.

Actuator limits. Rated voltage and overdrive clamp are set from the Adafruit 1201 datasheet before the first auto-calibration, starting conservatively. The 1201 is a 2–5 V part; 3.0 V rated with a 3.3 V clamp is the current assumption in firmware, to be confirmed at bring-up. Wrong values here produce a weak or buzzy actuator.

Why PWM, and what it costs

Gained	Given up
No bus transaction while the patient is moving, which is when connector and cable flex on a worn sleeve make the I^2C link least reliable.	The built-in effects library, which is selected by number over I^2C. Every pattern must instead be shaped by hand. For the six short patterns specified here this is a small amount of code and gives exact control of pulse widths and gaps.
Exact pulse timing. The specified patterns use 70–100 ms pulses with 80 ms gaps, which the library's fixed envelopes do not reproduce precisely.	One GPIO and one trace.
Driver behaviour is retained: overdrive and active braking still operate from the stored calibration in PWM input mode.	A 20 kHz switching signal routed near a microvolt-level analogue front end. Mitigated because drive is active only during a cue of a few hundred milliseconds, and cues fire when the patient is stationary at the confirming hold.

The practical conclusion is that PWM costs little and removes bus traffic from the part of the session where the link is most exposed. A boot-time I^2C failure remains possible, but it occurs while the patient is stationary, it is detectable by the register read-back, and the step can be retried; a failure mid-movement cannot be handled so cleanly. The effects library remains available as a fallback should PWM shaping prove troublesome in bring-up. The driver's EN pin is held by a GPIO, providing a hard mute independent of the bus; if EN is used to sleep the driver between cues, the configuration sequence is re-sent unconditionally on wake.

5.8 Signal chain

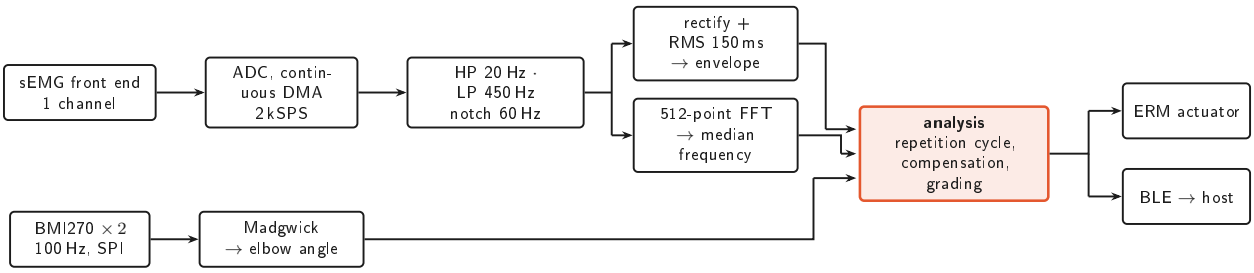


Figure 4: *Signal path. The haptic branch is independent of the BLE branch, so feedback continues when the host connection is lost.*

Two choices in this chain are load-bearing. The front end delivers raw band-limited sEMG rather than a rectified envelope, because envelope-only output discards the frequency content that median-frequency fatigue tracking requires. And acquisition runs in continuous DMA mode rather than one-shot conversions from a timer callback, because the jitter introduced by the latter degrades the spectral estimate that fatigue detection depends on.

Fatigue detection. Median frequency is computed from a 512-sample Hann-windowed FFT with 50% overlap, only while the activation envelope exceeds threshold, and tracked as a linear regression across the set. Three consecutive repetitions with a slope below the calibrated threshold raise a fatigue warning and trigger the rest cue, once per set. Settled on 22 October: the warning **advises and does not end the set**. The cue fires, the event is reported, and what to do about it is the patient's and the therapist's decision — a device that ended a set on its own reading of a frequency

trend would be overruling the person doing the exercise.

5.9 Host software

The device is authoritative for every event it reports — repetition counted, compensation, fatigue, fault — and the host displays those events rather than re-deriving them from the raw stream.

Phase 1 host: Python. A `bleak`-based client on a laptop serves four functions: issuing control commands, plotting the live feature stream, logging every received frame to CSV, and publishing repetition records to the same MQTT topics the phone application will use, so the backend is exercised before the phone application exists. The log is the input to offline tuning of filter coefficients, thresholds, and grading weights, which is required work regardless of host platform.

Conditions attached to this substitution: the client uses the exact BLE layout agreed for the phone application, so the phone client can replace it with no firmware change; the 29 October demonstration is scoped to the live view rather than the patient-facing game, which Python cannot exercise; and phone work continues in parallel against the same specification.

Implementation notes: the notification callback must do nothing but push into an `asyncio.Queue`, with parsing and plotting in a separate task, or the BLE connection stalls behind the plot; frames are capped at 180 bytes rather than the 244 a 247-byte MTU permits, because macOS caps ATT MTU at 185 and `bleak` does not expose a portable MTU request; at approximately 10.4 kB/s the raw stream is well within the limits of every backend. Passkey pairing behaves differently on macOS, so the passkey is disabled in prototype firmware builds.

Phase 2 host: Flutter application. A single Flutter codebase compiles to native Android and iOS builds, so both platforms are reachable from one body of Dart code with a cross-platform BLE package. Android is the development target, since deploying to an iOS device additionally requires macOS tooling and a paid Apple developer account; iOS is added from the same codebase once the Android build is stable. What the Python client cannot test — patient-facing user experience, the game, and platform BLE behaviour such as connection interval, background operation, and permissions — remains phone-side work.

Pairing and connection loss. The device advertises when idle, pairs once using a six-digit passkey printed on the pod, and reconnects automatically thereafter. Losing the link does not change the device state: acquisition, repetition detection, grading, and all haptic feedback continue, and only the transmit task idles. Repetition events are buffered to flash so they survive a power loss; the full-rate raw stream is buffered in PSRAM. The host recovers the missed range by sequence number on reconnection, and the device reports explicitly if any data was genuinely lost.

5.9.1 Parameters settable from the host.

All parameters below are runtime-configurable over BLE, may be changed during a session, take effect from the next repetition, and are stored in NVS and read back for confirmation.

Parameter	Default	Purpose
`hold_duration_ms`	1000 ms (300–3000)	Duration of the confirming hold at the extended position. The primary tuning control during volunteer testing.
`rep_target`	20 (1–50)	Repetitions per set; determines when the set-complete pattern is issued.
`set_count`, `rest_seconds`	3, 60 s	Session structure.
`target_angle`	110°	Therapist-set; the peak criterion is 80% of this value.
`start_angle`, `finish_angle`	15°, 10°	Raised together for patients who cannot fully extend.
`tempo_min`, `tempo_max`	1.5 s, 3.5 s	Prescribed tempo window.
Compensation thresholds	20°, 10°, 30°/s	Shoulder, trunk, and momentum sensitivity.
Grading weights and deductions	Equal weights, §5.5	Tuned against volunteer data.
`raw_stream_enable`	On during tuning	Switches the full-rate raw stream without affecting results or events.
Haptic pattern test	—	Fires a named pattern on demand for bench verification.

6 Performance metrics and acceptance criteria

Metric	Acceptance test
Joint angle accuracy	Within 5–10° of a goniometer at 0°, 45°, 90°, 135°
Calibration robustness	Each step repeatable individually; movement during steps 1–2 rejected; a hyperextended step 2 does not prevent repetitions from completing; the whole protocol completes in under 30 s
Repetition counting	Ten deliberate repetitions counted as exactly ten, on three consecutive runs; a passively swung arm produces no count. **[21 Oct test]** Both pass in `test/test_rep.c`
Partial rejection	A repetition abandoned mid-range, and one exceeding the 10 s timeout, both reported as partial and not counted
Feedback latency	Haptic delivered within 50 ms of the confirming hold completing
Pattern discrimination	A blindfolded team member identifies all seven patterns correctly 10/10 times, through fabric, and never mistakes the soft click for a result
Pattern timing accuracy	Pulse widths and gaps within 10% of specification, measured on the PWM line (proposed tolerance)
Driver configuration integrity	Zero register read-back mismatches across 50 consecutive power cycles (proposed)
No cue overlap	No two patterns drive the actuator at once; covered by host-side tests and a bench capture
Link independence	Cues continue with the BLE link disconnected mid-set
Cue interference	No measurable increase in the EMG noise floor or IMU artefacts attributable to a cue (proposed)
Compensation detection	Staged shoulder elevation flagged, the repetition still counted, and the score reduced
Contact-fault detection	A deliberately lifted electrode raises the fault cue with its specific cause
Fatigue detection	Median-frequency trend falls measurably across a deliberately fatiguing set, the rest cue fires once, and the set continues
Set completion	Set-complete pattern fires on the repetition reaching `rep_target`, for targets of 5 and 20
Parameter configuration	`hold_duration_ms` and `rep_target` changed from the host take effect on the next repetition and survive a power cycle
Raw stream control	Raw stream disabled and re-enabled mid-session without interrupting results or events
Link robustness	60 s logged with zero sequence gaps; a forced disconnection loses no data after recovery
Signal quality	Clipping counter reads zero at maximum effort once front-end gain is set
Battery life	Two hours of continuous operation; below 15% the device completes the set, then stops and saves

7 Milestones

Aligned to the course schedule. Weekly lab notebook entries are published to the project website by noon each Friday, and the team meets the instructor and TAs during the Thursday lab.

Date	Course deliverable	Project milestone
Oct 22	PRD on website, noon	This document; design forks closed
Oct 23–28	—	EMG channel live; DSP chain verified; both IMUs on SPI; elbow angle against a goniometer; all seven haptic patterns verified on a scope and through fabric; BLE stream logged
Oct 29	Rapid prototype on website, noon	Phase 1 demonstration: full signal path on the development board with MyoWare
Oct 30–Nov 2	—	PCB schematic complete and reviewed
Nov 3	Status presentation	10-minute update: goals, architecture, state, remaining risks
Nov 4–7	—	PCB ordered; volunteer data collection for the reference corridor begins
Nov 10	Peer review 1, noon	—
Nov 11–21	—	Grading tuned against volunteer data; sleeve and pod mechanical design; battery and charging
Nov 24–Dec 1	—	PCB bring-up; integration onto the sleeve
Dec 2–8	—	Full-system integration; two complete demo rehearsals
Dec 10	Final demo and presentation	Live demonstration per §3.1
Dec 15	Final report, peer review 2, noon	ACM double-column technical report

8 Team responsibilities

Member	Primary areas
Qian (Andy) Cai	Hardware: schematic capture, PCB layout, component selection, board bring-up, power and charging. Owns the development boards.
Aidan Goodman	Firmware architecture: acquisition layer, DSP chain, BLE service and the host interface contract, test suites.
Rusha Maiti	Algorithms and validation: calibration protocol, repetition detection and grading parameters, prototype testing and volunteer data collection.
Eyerusalem Earl	Host software and integration: Python client, Flutter application, backend topics, mechanical integration of sleeve and pod.
Rotating duties	Schedule management, weekly lab notebook, purchasing and reimbursement forms, website upkeep.

Assignments are primary areas rather than exclusive ownership; integration work is shared. *To be confirmed by the team at the 22 October lab and corrected here if different.*

9 Materials, budget, and outside help

Item	Status	Cost ($)
ESP32-S3 development boards (×3)	Owned	0
SparkFun BMI270 IMU breakouts (×2)	Ordered	37.00
Adafruit DRV2605L haptic drivers (×2)	Ordered	15.90
Adafruit 1201 ERM actuators (×5) — the selected actuator	Ordered	9.75
MyoWare 2.0 sensor, cable shield, electrodes, electrode cable	Ordered	65.15
Subtotal, Phase 1		*127.80*
Second MyoWare channel — no longer needed; the custom PCB carries both	*Not ordered*	*0*
Custom PCB fabrication and assembly	To order	150–250
PCB components (front end, ADS131M02, BMI270, DRV2605L, regulators, passives)	To order	∼80
LiPo cell, charge circuit, fuel gauge	To order	∼25
Ag/AgCl electrodes, consumable stock	To order	∼30
Compression sleeve, fabric, fasteners, 3D-printed pod	To order	∼40
Projected total		**∼510–610**

Figures beyond the Phase 1 subtotal are estimates and will be revised as quotations arrive. Purchases are reimbursed through the course form.

Two decisions of 22 October reduce the outstanding spend: the ERM units already purchased are the production actuator, so no LRA order is needed, and the single-channel prototype removes the second MyoWare module. Both were in the Phase 1 "to order" column in v1.4.

Facilities and tools. Lab bench in CSE 003E; oscilloscope and bench supply; Altium Designer for PCB design; a 3D printer for the pod and electrode housings; a goniometer for angle validation.

Outside help needed.

- A physical or occupational therapist, for review of the exercise protocol, realistic target angles and tempo ranges for patients, and what a clinician would want to see reported.
- Healthy volunteers for the reference corridor and for staged compensation testing.
- PCB fabrication and assembly service, per the course instructions.

10 Risks and mitigations

10.1 Project risks

Risk	Impact	Mitigation
PCB fabrication or assembly delay	High	Order by the first week of November; the Phase 1 development-board build remains a complete working fallback for the final demonstration
Custom analogue front end underperforms the MyoWare reference	High	Bring up the PCB front end against the MyoWare signal as a reference; keep the MyoWare path wirable
Triceps channel arrives only with the PCB	Medium	Co-contraction is unscored until then, and the donning check cannot detect a reversed sleeve. Both paths are written and guarded on one constant, so the PCB enables them rather than requiring new work
Electrode contact instability on a moving sleeve	High	Continuous contact monitoring with a fault cue; short leads; compression and foam backing; recalibration prompt
Component availability	Medium	Order early; select parts with second sources where possible
Volunteer availability for data collection	Medium	Recruit within the course cohort; begin collection in early November
Scope creep from the phone application and game	Medium	Phase 1 scoped to the live view; phone work proceeds in parallel and is not on the critical path
Actuator rated voltage or overdrive clamp set wrongly	High	Set from the Adafruit 1201 datasheet before the first auto-calibration; start conservatively
Driver configuration fails or is lost (boot-time I^2C error, or an EN sleep resetting the chip)	High	Register read-back self-test; configuration re-sent unconditionally on wake; stored-calibration fallback
Patterns indistinguishable through fabric	High	Distinguish by rhythm and length rather than intensity; test through fabric early; tune with volunteers
ERM spin-up smears the shortest pulses (the 40 ms tick and the 30 ms soft click)	Medium	Overdrive kick and active braking from the stored calibration; verify on a scope; lengthen the cue if needed
Haptic timing stretched by radio activity	Medium	The haptic task shares core 0 with the BLE host. Measure pulse widths on the PWM line with BLE active; if marginal, move segment timing to a hardware timer
20 kHz PWM couples into the EMG front end	Medium	Drive only during short cues at the stationary hold; route the trace away from the front end; compare the noise floor with and without cues
Cue vibration scored as an IMU jolt, or disturbing the next repetition	Medium	Cue blanking (§5.6), pending team agreement
Fault-state mute silences the fault cue	Medium	Play the fault cue to completion before cutting EN (§5.2)
Single points of knowledge in the team	Low	Weekly lab notebook; code and design files in the shared repository

10.2 Implementation hazards

Electrode contact drift during a session. [21 Oct test] The single largest source of bad data in the 21 October session. A sleeve or electrode shift after calibration raised resting EMG to roughly 40% of the calibrated maximum, invalidating the activation scale, the co-contraction score, and the

"relaxed between repetitions" check simultaneously.

BMI270 configuration upload. The BMI270 ships inert and requires an approximately 8 kB configuration blob to be uploaded before it produces data. A device that reports the correct chip ID but returns only zeros typically indicates a configuration upload that failed silently. The initialisation routine verifies `INTERNAL_STATUS` = `0x01` after upload, and the self-test fails explicitly if it does not.

BMI270 SPI mode entry. The device powers up in I^2C mode regardless of board configuration. On the breakouts: ADR jumper fully open (this selects SPI), I^2C pull-up jumpers cut, and header pins mapped SCL→SCK, SDA→PICO, ADR→POCI, CS→CS. The driver then performs a dummy read of register `0x00` and allows CSB to rise, which latches SPI mode. Omitting this produces invalid data on all subsequent transfers.

Front-end gain and clipping. Clipping degrades the median-frequency estimate while leaving the activation envelope apparently plausible, so it is not detectable by inspection. The firmware maintains a per-channel clipping counter, reported in telemetry, and gain is set against that counter rather than visually. Calibration step 3 raises gain automatically when the maximum contraction reads too low.

Internal ADC noise in Phase 1. The ESP32-S3 internal ADC provides roughly 9.5 effective bits, against approximately 110 dB of dynamic range for the ADS131M02. Adequate for validating the chain, not for the product. If the signal appears unusable after filtering, investigate in this order: front-end gain, then electrode contact and reference placement, then ADC attenuation and calibration.

Yaw drift without a magnetometer. Absolute heading is unobservable with this sensor set, so elbow angle is always the relative rotation between the two IMUs with respect to their neutral-pose references. Mitigations: the per-session re-zero, and a rest-phase re-anchor to gravity whenever both IMUs report low angular velocity and acceleration near 1 g for more than 300 ms — which the confirming hold now guarantees once per repetition.

Gyroscope bias corruption at calibration. **[21 Oct test]** Movement during the rest step corrupts gyroscope bias and therefore the joint angle for the whole session, with no obvious symptom. The stillness gate exists to prevent this.

PSRAM configuration. `CONFIG_SPIRAM_MODE_OCT` must be set; the default quad mode causes an N8R8 module to report no PSRAM. The session buffer is allocated with `heap_caps_malloc(size, MALLOC_CAP_SPIRAM)` and verified at boot.

Buffer sizing. At 2 kSPS across two channels plus IMU data the raw stream is approximately 648 kB per minute, while only 300–350 kB of internal SRAM remains free once the BLE stack is loaded. Buffering is tiered: DMA landing buffers, an 8-second live ring in SRAM, and the full session in PSRAM. Repetition events are additionally written to flash so they survive power loss.

Mains interference. Battery operation limits 60 Hz pickup, but lead length makes the notch filter necessary, and the setup contact check measures pickup explicitly. A 180 Hz harmonic notch sits behind a compile-time flag.

Trunk lean is inferred. Detected as common-mode rotation of the two arm IMUs rather than measured. A trunk-mounted IMU would be required to measure it, and is not in scope.

11 Out of scope for this quarter

- **The patient-facing game.** Development may proceed in parallel, but it is not demonstrated and the Python client cannot exercise it.
- **The therapist dashboard.** The device-side data path and the MQTT topics are in place and exercised from the Python client; the dashboard itself is later work.
- **Remote adjustment of exercise targets from a clinician portal.** Parameters are adjustable from the local host only.
- **Wi-Fi connectivity and an on-device web interface.** BLE is the single host transport.
- **Machine-learning models.** Repetition detection uses deterministic thresholds, and no labelled dataset yet exists — the current thresholds are what will generate the labelled boundaries a model would require. Learned quality scoring is the natural second version.
- **Dry and textile electrode evaluation.** Snap Ag/AgCl only.
- **Multiple patient profiles per device.**
- **Wrist-substitution detection.** Not observable with this sensor set; it requires a third IMU on the hand.
- **Clinical validation and any diagnostic claim.** See §13.

12 Decisions and open questions

12.1 Decisions settled on 22 October

The four below were taken after the firmware had been brought into line with v1.4, and each reverses or closes something that document left open.

Question	Decision
Prototype channel count	**One** EMG channel, on the biceps. Three electrodes. The triceps channel and the co-contraction score arrive with the custom PCB, which carries both without a second module or a second reference.
Actuator	**ERM** for the full production cycle. The v1.4 move to an LRA is reversed: the ERM units are already purchased, the DRV2605L's overdrive and braking make the specified rhythms workable, and nothing in the pattern vocabulary depends on resonance tracking.
Soft click at 80% of target	**Yes.** Added as a seventh pattern, deliberately shaped so it cannot be mistaken for a result (§5.6).
Fatigue warning	**Advises only.** The cue fires and the event is reported; the set continues. A device that ended a set on its own reading of a frequency trend would be overruling the person doing the exercise.

12.2 Decisions carried forward from v1.4

Question	Decision
Haptic drive path	I^2C configuration at power-up, switching to PWM drive immediately and for the remainder of the session. Rationale and costs in §5.7.1.
Where a repetition is counted	On return to the extended position, after a configurable confirming hold. Tempo therefore covers the full excursion, and the hold provides the stationary moment that feedback and gravity re-anchoring both need.
Tempo versus hold steadiness	Tempo is scored; hold steadiness is computed and reported but is not part of the default score.
Status indicator	RGB LED on the custom PCB, giving colour-coded states as well as blink patterns.
Phone platform	Flutter, which compiles to both Android and iOS from one codebase. Android first, iOS added from the same source.
Phase 1 sensing	MyoWare 2.0 for the 29 October prototype; the custom front end follows on the PCB.
Haptic timing	Patterns shaped by hand over PWM rather than selected from the driver's effects ROM, for exact control of pulse widths and gaps.
Raw data stream	Sent at full rate but switchable, so tuning sessions record it and ordinary use does not.
Data lost to a dropout	Replayed by sequence number on reconnection.
Flagged repetitions	Count toward the repetition target; quality is expressed in the score.
Live parameter changes	All listed parameters may be changed mid-session, taking effect from the next repetition.

12.3 Still open

By	Question	Blocks
Oct 23	Adafruit 1201 rated voltage and overdrive clamp from the datasheet	Driver auto-calibration values. Firmware assumes 3.0 V rated, 3.3 V clamp until confirmed
Oct 24	Agree cue blanking: suspend IMU and EMG analysis for the duration of each cue?	Scoring behaviour around cues; firmware and algorithms owners both affected. The soft click makes this slightly more pressing, since it is the one cue that fires while the patient is moving
Oct 25	Electrode sites confirmed on the biceps belly	Volunteer data collection; not a firmware-structure dependency
Oct 25	Shared-reference front end verified on the bench against separate references	Five-electrode layout; PCB schematic
Oct 25	Haptic pattern timing measured on the PWM line with BLE active	Whether segment timing can stay on the scheduler or must move to a hardware timer
Nov	Demonstration covers elbow flexion only, or adds wrist flexion and extension?	Electrode siting; sleeve design
Nov	Battery capacity, fuel gauge part, and charge circuit against the two-hour target	Pod design

12.4 The BLE interface, in plain terms

Over BLE the device advertises a set of named channels, and the host subscribes to the ones it needs. Each channel has an identifier and a defined message layout. The identifiers themselves are arbitrary — any value works, provided firmware and host use the same one — so they are not a design decision; they are generated once and recorded in a shared header file. The decisions that matter are what the device sends, how often, and what the host may change.

Channel	Content and rate
Commands (host → device)	Start, stop, run a named calibration step, set parameters, read parameters, test a haptic pattern, enable or disable the raw stream, set clock, power off
Live features (device → host)	Envelope, joint angle, state, repetition count and whether the agonist is active, at 20 Hz on air; analysis runs internally at 50 Hz
Results (device → host)	Per-repetition score and flags, plus events: partial repetition, compensation, fatigue, contact fault, set complete
Raw data (device → host)	Full-rate EMG and IMU samples, ∼10.4 kB/s, switchable
Status (device → host)	State, battery, signal quality, and the range of data the device still holds
Recovery (device → host)	Replay of data missed during a disconnection, requested by range, served below the live stream in priority
Configuration (device → host)	The parameters of §5.9.1 as the device holds them, read back so the host can confirm a change took effect

All multi-byte values are little-endian, and every sample carries a timestamp from a single monotonic device clock. The host sends its own wall-clock time on connection; the device records the offset and reports it, but never adjusts its own clock, since altering the sampling clock would corrupt the spectral analysis.

13 Changes from the October project proposal

Proposal	Current	Reason
MyoWare 2.0 and ADC 15 Click as the final front end	One MyoWare for Phase 1; integrated two-channel front end with ADS131M02 on the PCB	Single integrated board. Three electrodes for the prototype; five on the PCB, which adds the triceps pair and shares one driven reference
Single maximum-contraction step	Separate biceps and triceps maxima, the triceps deferred to Phase 2	**[21 Oct test]** The co-contraction score normalises the triceps and needs its own reference; with one channel there is nothing to normalise yet
90 s calibration	Six steps, ∼19 s of capture, individually repeatable	A failed step should not require restarting the protocol
Haptic within 200 ms of the compensating movement	Feedback within 50 ms of the confirming hold completing	Feedback is scheduled rather than reactive
Two IMUs on I^2C	Two IMUs on SPI	No sensor data on a shared bus
Phone to cloud to therapist dashboard	Python host publishing to the same MQTT topics; Flutter application in parallel; dashboard later	Logging is required for tuning regardless, and the backend is exercised early
Haptic patterns unspecified	Seven defined patterns, with distinct fault and set-complete cues, a mid-movement soft click, and explicit priority and preemption rules	**[21 Oct test]** The original double-pulse scheme collided with the fault cue; patterns are distinguished by rhythm
ERM actuator	**ERM retained**	An LRA was proposed in v1.4 and reversed on 22 October: the units are purchased, and the DRV2605L's overdrive and braking already make the specified rhythms workable
Fixed repetition protocol	Hold duration, repetition target, thresholds, tempo window, and grading weights configurable from the host	Required for tuning against volunteer data without rebuilding firmware

14 Safety, ethics, and standards

RehabSense is a research prototype and makes no diagnostic claim. It is battery powered and is not connected to mains-powered equipment while worn; the analogue rail is cut in hardware whenever the device is charging. Testing is limited to team members and consenting healthy volunteers; validation with stroke patients is future work and outside the scope of this course. Session data is health-related, and the current design keeps raw data on the device and on the local host machine.

Applicable standards. Were RehabSense developed as a product rather than a course prototype, the governing standard would be **IEC 60601-1** for the basic safety and essential performance

of medical electrical equipment, together with **IEC 60601-1-11** for equipment used in the home healthcare environment. The relevant requirement for this design is patient isolation: the electrodes are in direct skin contact, so leakage current limits and the prohibition on a mains path to the patient apply directly, which is why the device is battery powered and the analogue rail is disconnected during charging. Classification under the FDA framework would likely place a device of this kind as a Class II powered muscle stimulator-adjacent or biofeedback device requiring 510(k) clearance; the project makes no such claim and is not submitted for it.

A Related research (EE 542)

Cifrek et al., "Surface EMG based muscle fatigue evaluation in biomechanics" (2009) [3]. This review surveys methods for detecting localised muscular fatigue from surface EMG, with emphasis on spectral indices: as a muscle fatigues, the power spectrum of its EMG shifts toward lower frequencies, and the mean and median frequency of that spectrum fall measurably over a sustained contraction. The paper also catalogues the practical difficulties — non-stationarity of the signal during dynamic contractions, sensitivity to electrode placement, and the need for adequate bandwidth and sampling. Its influence on this project is direct and structural. It is the reason the system samples raw, band-limited EMG at 2 kSPS per channel rather than the pre-rectified envelope a sensor module can supply: the envelope discards exactly the spectral content the fatigue index depends on. It is also why median frequency is computed only while the activation envelope exceeds threshold and is tracked as a regression across a set rather than evaluated on a single window, and why clipping is counted and reported rather than assumed absent.

Hermens et al., "Development of recommendations for SEMG sensors and sensor placement procedures" (2000) [4]. This paper reports the SENIAM project's consensus recommendations for surface EMG sensor design and placement, including bipolar electrode configurations, an inter-electrode distance on the order of 20 mm, placement over the muscle belly aligned with the fibre direction, and reference placement on electrically inactive tissue. Its influence on this project appears in the electrode layout: two electrodes at approximately 20 mm spacing along the belly of the biceps, with a reference on the bony prominence at the elbow, and the triceps pair added on the same pattern when the custom PCB arrives. It also motivates the donning check, since the recommendations only hold if the sensors are actually where the protocol assumes.

With two channels that check is a comparison — the biceps channel must exceed the triceps during a prompted curl, which catches a reversed sleeve and says so specifically. With one channel there is nothing to compare against, so the prototype instead requires the envelope to rise while the elbow flexes: enough to catch a lifted, misplaced or dead electrode, not enough to name a reversed sleeve, and the message the patient sees says so rather than claiming more. Finally, it is the basis for treating electrode siting as a fixed, documented procedure illustrated in the application rather than something left to the patient's judgement.

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